

2.7 Trace data retrieval in Array, List and Queue

ایرے میں اگر انڈیکس معلوم ہو تو سیدھی چھلانگ لگتی ہے، جیسے $\text{Array}[2] = 20$.

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 16 of 16

2.7 Trace data retrieval in Array, List and Queue

LEARN IT

Tracing data retrieval means following the step-by-step path of how an algorithm finds or accesses data. Array, linked list and queue are all linear, but the **effort** to find an item is very different because of how they are organised inside.

2.7.1 Retrieval in an array

Every element has a fixed index from 0 to $n-1$ pointing at a spot in memory. If the algorithm already knows the index, it does not search: it **jumps** (direct access, $O(1)$).

Array = [10, 50, 20, 70, 90, 30]. Direct access: $\text{Array}[2] = 20$.

Loop iteration	Index	Array access	Value
1	0	Array[0]	10
2	1	Array[1]	50
3	2	Array[2]	20

```
Array = [10, 50, 20, 70, 90, 30]
print("Direct Array[2] =", Array[2])
for i in range(3):
    print(i + 1, i, f"Array[{i}]", Array[i])
```

OUTPUT

```
Direct Array[2] = 20
1 0 Array[0] 10
2 1 Array[1] 50
3 2 Array[2] 20
```

2.7.2 Retrieval in a linked list

Linked lists have **no index**. The only way is to start at HEAD and follow links one node at a time (sequential access, $O(n)$).

List: [12] → [98] → [23] → [87] → [56] → NULL. Search key: 87.

Step	Node data	Compare with 87	Action
1	12	12 != 87	Move next
2	98	98 != 87	Move next
3	23	23 != 87	Move next
4	87	87 == 87	FOUND — output True

```

nodes = [12, 98, 23, 87, 56]
key = 87
found = False
for i, value in enumerate(nodes, start=1):
    print(f"Step {i}: Check Node {i} ({value}).")
    if value == key:
        found = True
        print("FOUND! Output: True")
        break
    print("Not equal. Move Next.")
print("Result:", found)

```

OUTPUT

```

Step 1: Check Node 1 (12).
Not equal. Move Next.
Step 2: Check Node 2 (98).
Not equal. Move Next.
Step 3: Check Node 3 (23).
Not equal. Move Next.
Step 4: Check Node 4 (87).
FOUND! Output: True
Result: True

```

Key 87 sits at the fourth node, so you paid four checks. An array that already knew index 3 would have paid one jump.

2.7.3 Retrieval in a queue

A queue is FIFO. You **cannot** retrieve from the middle or the rear. The oldest item at the Front is always DEQUEUEEd first.



You cannot pull B until A has left.

Queue A, B, C. ENQUEUE D joins the rear. DEQUEUE must remove A, not C or B.

```
from collections import deque
q = deque(["A", "B", "C"])
q.append("D")
print("after ENQUEUE D:", list(q))
print("DEQUEUE:", q.popleft())
print("now:", list(q))
```

OUTPUT

```
after ENQUEUE D: ['A', 'B', 'C', 'D']
DEQUEUE: A
now: ['B', 'C', 'D']
```

WORKED EXAMPLE

Worked example: three retrievals compared for "get 20 / get 87 / get A".

Structure	Target	Path	Can you jump?
Array [10, 50, 20, ...]	20 at index 2	Array[2] in one step, or scan 10, 50, 20	Yes, if the index is known
Linked list 12→98→23→87→...	87	Must visit 12, 98, 23, then 87	No
Queue A-B-C	A (the only legal retrieval)	DEQUEUE Front	No — never the middle

REMEMBER IT

Array jumps, list walks, queue waits. If the exam asks "why is array retrieval different?", write **direct access by index** versus **sequential links** versus **Front only**.

COMMON MISTAKE

Mistake: DEQUEUEing C because "C is the interesting letter", or reading linked-list node 4 as `list[3]` without walking.

Fix: show every hop in the table. Marks are for the path, not only for FOUND.

اردو میں سمجھیں

ایرے میں اگر انڈیکس معلوم ہو تو سیدھی چھلانگ لگتی ہے، جیسے `Array[2] = 20`۔ لنکڈ لسٹ میں HEAD سے ایک ایک نوڈ دیکھنا پڑتا ہے؛ 87 چوتھے نوڈ پر ملا۔ کیوں صرف Front نکلتا ہے، درمیان سے نہیں؛ A, B, C میں DEQUEUE ہمیشہ A ہے۔

CHECK YOURSELF

1. What is Array[2] in [10, 50, 20, 70, 90, 30]?
2. How many nodes are checked before 87 is found in $12 \rightarrow 98 \rightarrow 23 \rightarrow 87 \rightarrow 56$?
3. In queue A, B, C, can you retrieve B without removing A?

Answers: 1. 20. 2. Four (12, 98, 23, 87). 3. No — a queue only DEQUEUEs the Front, so A must leave first.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Tracing retrieval means:

- a. Following how an algorithm finds/returns an item
- b. Deleting the DS
- c. Drawing Figma
- d. Running pip

MCQ 2. Array retrieval by index is:

- a. Direct: $a[i]$
- b. Walk from head always
- c. Pop until empty always
- d. BFS always

MCQ 3. Linked-list retrieval of the k-th is:

- a. Walk k links
- b. $O(1)$ index
- c. A binary cut
- d. A hash

MCQ 4. Queue retrieval of the next client is:

- a. Dequeue front (FIFO)
- b. Pop top (LIFO)
- c. Random node
- d. Tree root

MCQ 5. You cannot take the middle of a queue without:

- a. Breaking FIFO (or using another DS)
- b. It is the definition of queue
- c. A pie
- d. A sprint

MCQ 6. A trace should show:

- a. Pointers/indexes before and after
- b. Only the career names
- c. Only colours
- d. Only OSI

Extra short questions

1. Trace getting $a[2]$ from array $[10,20,30,40]$.
2. Trace finding 30 in a list $10 \rightarrow 20 \rightarrow 30$.
3. Trace queue A-B-C serve one.
4. Why tracing retrieval is in the syllabus.

Extra long question

1. Same three names stored in array, list, queue. Trace "get the second person" in each. Who is not allowed in a pure queue?

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Index 2 $\rightarrow$ 30 in one step.
2. Start 10 $\neq$, next 20 $\neq$, next 30 $=$. Three visits.
3. Dequeue A; front is B. Remaining B-C.
4. So you can justify the DS with actual steps, not slogans.

Long

1. Array `a[1]`; list two hops; queue: you may only legally take the front — getting the second requires dequeue then the new front, which also removes the first. That is the point.

Next lecture: Chapter exam (in the chapter lecture PDF)